ANIMAL CELL TECHNOLOGY: PRIMARY CELL ISOLATION FROM SHRIMP HEPATOPANCREAS

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Abstract

This study presents a standardized laboratory workflow for the isolation and establishment of a primary cell suspension from shrimp hepatopancreatic tissue. The procedure integrated mechanical mincing with enzymatic dissociation using 0.25% Trypsin in Leibovitz L-15 medium, followed by sequential filtration through 100 μm and 40 μm mesh to ensure optimal single-cell recovery. The resulting suspension was concentrated via centrifugation at 500g for 10 minutes. Cell viability and density were subsequently quantified using the Trypan blue exclusion assay in a 1:1 dilution ratio via hemocytometer analysis. The protocol successfully yielded a viable primary cell suspension, demonstrating that the synergy of mechanical and enzymatic treatments is an effective approach for harvesting crustacean cells. These findings provide a foundational methodology for downstream applications in marine biotechnology, pathobiology, and in vitro cellular assays.

Keywords: Shrimp; Primary cell culture; Hepatopancreas; Enzymatic dissociation; Trypan blue exclusion; Cell viability.

1. Introduction

2. Materials and Methods

2.1. Work flow

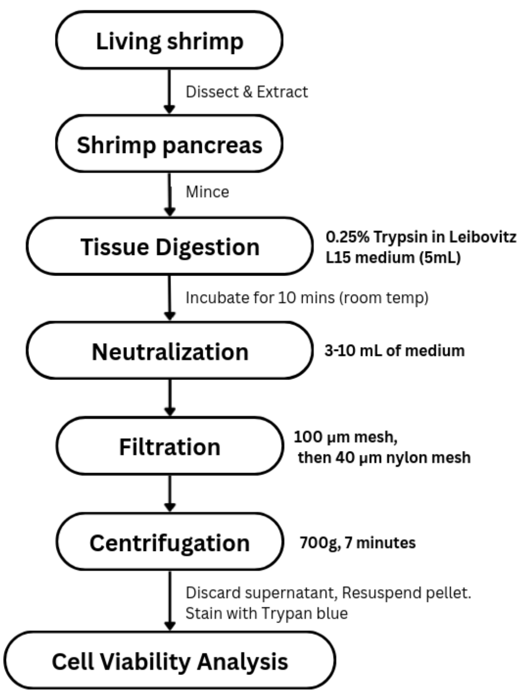

Figure 1: Workflow of primary cell isolation from shrimp hepatopancreas.

3. Results & Discussion

The cell isolation from shrimp hepatopancreas was quantified using a hemocytometer after a 1:1 dilution (10 μL cell suspension + 10 μL Trypan Blue). The resulting suspension was resuspended in 1 mL.

Table 1: Hemocytometer Counts and Viability Analysis

Square	Live (L)	Dead (D)	Total (T)	Viability (%)
Top	61	15	76	80.3%
Left	46	12	58	79.3%
Center	64	14	78	82.1%
Right	64	6	70	91.4%
Bottom	64	8	72	88.9%
Sum/Avg	299	55	354	84.5%

3.1. Calculations

1. **Percentage Viability:**

$$\%V = \frac{\sum L}{\sum T} \times 100 = \frac{299}{354} \times 100 \approx 84.5\%$$

2. **Live Cell Concentration:**
(DF = 2 due to 1:1 mix)

$$\text{Cells/mL} = \frac{\sum L}{5} \times DF \times 10^4 = 1.196 \times 10^6$$

3. **Total Yield (1 mL):**

$$\text{Yield} = 1.196 \times 10^6 \text{ viable cells}$$

The data demonstrates a successful isolation with a viability of 84.5%, exceeding the 80% benchmark for healthy primary cultures. The 0.25% Trypsin treatment (10 min) effectively dissociated the tissue without significant membrane damage. The concentration ($\approx 1.2 \times 10^6$ cells/mL) indicates a high recovery rate. Minor localized viability decreases (Top/Left) are negligible compared to the robust overall population.

4. Conclusion

The protocol involving trypsinization, sequential filtration ($100 \rightarrow 40$ μm), and resuspension yielded a high-density, viable cell suspension (1.196×10^6 cells, 84.5% viability), optimized for downstream applications.

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